

Questions with Answer Keys

MathonGo

Q1 - 2024 (27 Jan Shift 2)

Considering only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x) + \tan^{-1}(2x) = \frac{\pi}{4}$ is :

- (1) More than 2
- (2) 1
- (3) 2
- (4) 0

Q2 - 2024 (29 Jan Shift 2)

Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos(2 \sin^{-1} x) = \frac{1}{9}$ and let α, β ($\alpha > \beta$) be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point (α, β) lies on the line

- (1) $3x + 2y = 2$
- (2) $5x - 8y = -9$
- (3) $3x - 2y = -2$
- (4) $5x + 8y = 9$

Q3 - 2024 (31 Jan Shift 1)

For $\alpha, \beta, \gamma \neq 0$. If $\sin^{-1} \alpha + \sin^{-1} \beta + \sin^{-1} \gamma = \pi$ and $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$, then γ equal to

- (1) $\frac{\sqrt{3}}{2}$
- (2) $\frac{1}{\sqrt{2}}$
- (3) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
- (4) $\sqrt{3}$

Q4 - 2024 (31 Jan Shift 2)

If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to

- (1) $4\pi^2 + 25$
- (2) $8\pi^2 - 40\pi + 50$

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(3) $4\pi^2 - 20\pi + 50$

(4) 25

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Solutions

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Q1

$$\tan^{-1} x + \tan^{-1} 2x = \frac{\pi}{4}; x > 0$$

$$\Rightarrow \tan^{-1} 2x = \frac{\pi}{4} - \tan^{-1} x$$

Taking tan both sides

$$\Rightarrow 2x = \frac{1-x}{1+x}$$

$$\Rightarrow 2x^2 + 3x - 1 = 0$$

$$x = \frac{-3 \pm \sqrt{9+8}}{8} = \frac{-3 \pm \sqrt{17}}{8}$$

$$\text{Only possible } x = \frac{-3+\sqrt{17}}{8}$$

Q2

$$\text{Assume } \sin^{-1} x = \theta$$

$$\cos(2\theta) = \frac{1}{9}$$

$$\sin \theta = \pm \frac{2}{3}$$

as m and n are co-prime natural numbers,

$$x = \frac{2}{3}$$

$$\text{i.e. } m = 2, n = 3$$

So, the quadratic equation becomes $2x^2 - 3x + 1 = 0$ whose roots are $\alpha = 1, \beta = \frac{1}{2}$ $\left(1, \frac{1}{2}\right)$ lies on

$$5x + 8y = 9$$

Q3

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Solutions

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Let $\sin^{-1} \alpha = A$, $\sin^{-1} \beta = B$, $\sin^{-1} \gamma = C$

$$A + B + C = \pi$$

$$(\alpha + \beta)^2 - \gamma^2 = 3\alpha\beta$$

$$\alpha^2 + \beta^2 - \gamma^2 = \alpha\beta$$

$$\frac{\alpha^2 + \beta^2 - \gamma^2}{2\alpha\beta} = \frac{1}{2}$$

$$\Rightarrow \cos C = \frac{1}{2}$$

$$\sin C = \gamma$$

$$\cos C = \sqrt{1 - \gamma^2} = \frac{1}{2}$$

$$\gamma = \frac{\sqrt{3}}{2}$$

Q4

$$a = \sin^{-1}(\sin 5) = 5 - 2\pi$$

$$\text{and } b = \cos^{-1}(\cos 5) = 2\pi - 5$$

$$\therefore a^2 + b^2 = (5 - 2\pi)^2 + (2\pi - 5)^2$$

$$= 8\pi^2 - 40\pi + 50$$

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